Prompt: React Course Filter & Timetable Builder

Build a single-page React application that lets a student browse a university course catalog, filter it every way possible, and assemble a personal weekly timetable with automatic conflict detection.

Data Source

The app is powered by a course catalog parsed from an Excel pre-registration sheet. Each course record has this shape:

```
{
  id: string,              // S.No.
  branch: string,          // e.g. "AE",
"CS", "EE"
  code: string,            // e.g. "AE201M"
  name: string,            // e.g.
"INTRODUCTION TO AEROSPACE ENGINEERING"
  prerequisite: string,    // free text,
may be empty
  slotType: string,        // e.g. "OE-1 DC
/ FIRST-HALF", "CORE 12 DC / REGULAR"
  credits: string,         // e.g. "3-0-0-
0(5)"
```

```
  instructor: string,      // may list
multiple names with (I)/(O) tags
  instructorEmail: string,
  sessions: [              // one entry per
distinct meeting time
    { days: ["M","W","F"], start: "17:00",
end: "18:00", venue: "EEM117" }
  ],
  fch: string               // "Not
Available" | "FCH Document" | link
}
```

Parse the raw sheet's time strings (e.g. `M (EEM117) W (EEM117) F (EEM117) 17:00-18:00`) into the `sessions` array — a course can have up to 3 separate session blocks with different days/venues.

Core Features

1. Filter Panel

Support simultaneous, combinable filters:

- **Branch/Department** (multi-select)
- **Course type/slot** (Core, Elective, Slot-1/2/3, OE, DC, etc. — multi-select)
- **Credits** (exact or range)
- **Instructor** (searchable dropdown)
- **Day of week** (multi-select)
- **Time range** (start/end sliders or pickers)

- **Venue** (multi-select)
- **Free-text search** across course code/name
- **Has prerequisite** toggle (yes/no/any)
- **Clear all filters** button
- Show live result count as filters are applied

2. Course List/Results

- Card or table view of filtered courses
- Each row shows code, name, branch, credits, instructor, timing summary
- "Add to Timetable" button per course
- Disable/gray out courses that conflict with the current timetable, with a tooltip explaining the conflict

3. Timetable Builder

- Weekly grid, Monday–Friday (or Mon–Sat if data requires), hourly rows
- Added courses render as colored blocks in their correct day/time cells, spanning multiple sessions if needed
- Click a block to remove it from the timetable
- **Automatic conflict detection**: overlapping time blocks are visually flagged (red border/background) and a warning banner lists the conflicting course pairs

- Running total of credits for the current timetable
- "Clear timetable" button

4. Persistence

- Save the current timetable and filter state so it survives a page refresh/reopen (use in-memory/local persistence appropriate to the environment — no browser localStorage if building as a Claude artifact; use the artifact storage API instead)

UI/UX Requirements

- Clean, responsive layout: filters in a collapsible sidebar (or top bar on mobile), results in the center, timetable as a separate tab/panel
- Use a consistent color per branch or course type for quick visual scanning in the timetable
- Empty states ("No courses match your filters") and loading states handled gracefully
- Keyboard-accessible filter controls

Tech Stack

- React functional components with hooks (`useState`, `useMemo`, `useEffect`)
- Tailwind utility classes for styling

- `lucide-react` for icons
- All logic in a single file/component tree, no external backend — data is loaded from a bundled JSON array parsed ahead of time from the source spreadsheet

Deliverable

A single React artifact/component (default export) that runs standalone, pre-loaded with the parsed course data, implementing all filters and the timetable builder described above.